

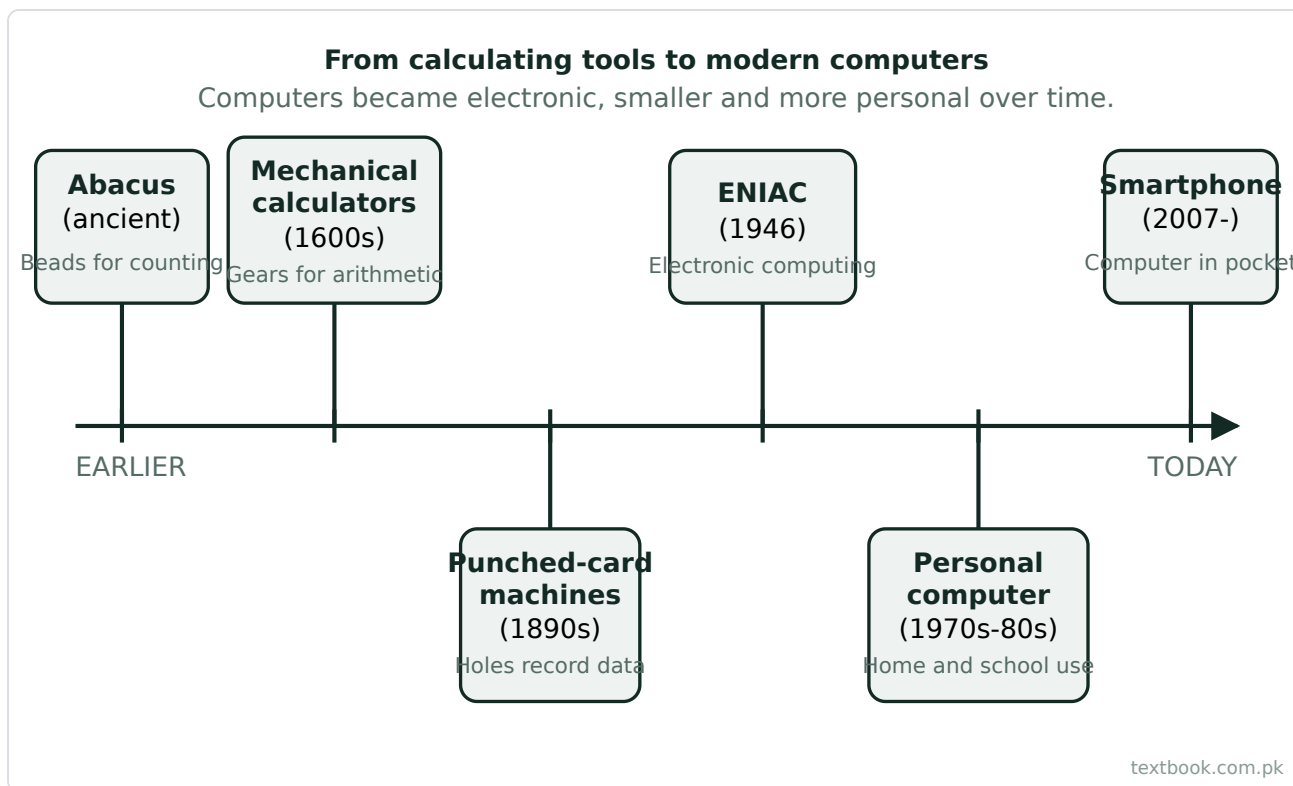
Draft - expert review pending

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Topic 1.2 - From calculating tools to modern computers

A real classroom situation

During a lesson on machines, a Class 7 pupil in a Karachi school raises his hand: "Sir, who made the first computer? Was it a scientist in America?" The teacher, Mr. Danial, starts to answer and then stops. He remembers that the word "computer" once meant a person: in the nineteenth century, a "computer" was a human being who computed, patiently adding columns of numbers for banks, insurance firms and observatories. The machine that took over the name is only about eight decades old, and the pocket-sized computer in the pupil's own bag is younger than the teacher. Mr. Danial decides the question deserves a proper answer, because the story of how calculating tools became computers explains what a computer actually is.



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Explanation

Evolution of computers: from early calculating devices to the modern computer

The story of the computer is the story of humans handing calculation work to machines, one step at a time.

The abacus, used for thousands of years across Asia including this region, let a trader add and subtract quickly by sliding beads along rods. It computed, but it computed nothing on its own: a person moved every bead.

In the 1600s and 1700s in Europe, mathematicians built mechanical calculators. Blaise Pascal's machine added and subtracted; Gottfried Leibniz's could also multiply and divide. Then, in the 1800s, two ideas changed everything. Joseph Jacquard's weaving loom controlled its patterns using punched cards, showing that a machine could follow instructions stored on a card. Charles Babbage designed his Analytical Engine, on paper, as a general calculating machine with a store (memory), a mill (processor) and punched-card input. The Analytical

Engine was never completed, but it is the first design we would recognise as a computer.

By 1890, Herman Hollerith's punched-card tabulating machines processed the United States census in years instead of decades, and the company he founded later became part of IBM. The first electronic general-purpose computer, ENIAC, was completed in 1946. It filled a large room, weighed about 27 tonnes and used thousands of vacuum tubes. The stored-program idea, published by John von Neumann's team in 1945, then let a computer hold its instructions in the same memory as its data, which is how every computer since has worked.

The next shrinking steps are the ones your pupils can see. The personal computer of the late 1970s and 1980s, such as the IBM PC and the Apple II, put a computer on a desk. The smartphone, from 2007 onwards, put one in a pocket, with more computing power than the room that held ENIAC.

A common misunderstanding says a computer is only the desktop machine on a desk. In truth, anything that takes input, processes it automatically and produces output is a computer: a smartphone, an ATM, a digital watch, even the controller inside a washing machine. The desktop shape is just one costume the computer wears.

Generations of computers: first to fifth

Computer historians divide the machine's history into five generations, each defined by the technology that built its switching circuits. The table below is the standard account given by introductory texts such as Shelly and Vermaat's *Discovering Computers* and Norton's *Introduction to Computers* (cited here at bibliographic level as the course guide lists them); Khan Academy's free course tells the same story for beginners.

Computer generations

Core technology and example machine

Generation	Years	Core technology	Example
First	1940s-1950s	vacuum tubes	ENIAC
Second	1950s-1960s	transistors	IBM 1401
Third	1960s-1970s	integrated circuits	IBM System/360
Fourth	1970s-present	microprocessors	IBM PC and Apple II
Fifth	present and future	artificial intelligence and very large scale integration	modern AI systems

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The pattern to notice is that each generation made computers smaller, cheaper, more reliable and faster at the same time. Some people assume each new generation is simply bigger and faster in every way. The opposite happened with size: the room-sized ENIAC shrank into a pocket-sized phone while gaining power a million times over. Speed grew; physical size fell.

Where does the school's equipment fit? An old desktop in a school office is a fourth-generation machine, built around a microprocessor. Modern phones, laptops and tablets are also fourth-generation in their construction, but they run the pattern-matching software we associate with the fifth generation: voice assistants, automatic translation and camera apps that recognise faces. The fifth generation is less a new kind of switch and more a new kind of use.

Activity: Generation gallery walk

Work in groups of three or four. About 20 minutes.

1. Your teacher posts five sheets around the room, one per generation, each naming only the core technology (vacuum tubes, transistors, integrated circuits, microprocessors, artificial intelligence).

2. Each group receives a card describing one machine: ENIAC, an IBM 1401 ledger machine, an IBM System/360, an early IBM PC, and a phone running a voice assistant.
3. As a group, decide which generation your card belongs to, write the machine's name on a sticky note, and place it on the matching sheet with one sentence of justification on the back.
4. Gallery walk: read the other groups' placements and note any you would challenge. Be ready to defend a placement using the generation's defining technology, not the machine's age alone.

Check your understanding

1. Name the core technology that defines each of the five computer generations. (*Remember*)
2. What did Jacquard's loom contribute to the history of computing? Answer in one sentence. (*Understand*)
3. A museum label reads: "Room-sized machine, 1946, thousands of vacuum tubes." Which generation is this machine from, and what single fact in the label proves it? (*Apply*)
4. Explain in two sentences why a smartphone is a computer even though it does not look like a desktop machine. (*Understand*)

Summary

Computers evolved from hand-driven calculating tools (the abacus, mechanical calculators) through instruction-following machines (punched cards, Babbage's design) to electronic computers: the room-sized ENIAC, the stored-program design, the desktop personal computer and the smartphone. The five generations are defined by their core technology: vacuum tubes, transistors, integrated circuits, microprocessors, and today's artificial-intelligence systems. A computer is anything that takes input, processes it automatically and produces output, whatever shape it wears.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the five computer generations and the technology that defines each.
- ☐ I can describe two milestones in the evolution of computers before 1946.
- ☐ I can classify a described machine into its generation using its defining technology.

Try this at your practicum school

Find the oldest piece of ICT equipment in your practicum school (an old desktop, a typewriter, a duplicator, a feature phone). Write three sentences about what it was used for, what replaced it if anything has, and which generation it belongs to if it is a computer.

Summative task

Prepare a one-page "computing family tree" for a Class 6 wall display. Choose five machines or devices from any period of computing history, place them on a simple timeline, and for each one write one line stating what it could do and one line naming its generation and defining technology if it is a computer.

Marking guide (10 marks):

- Five devices placed in correct time order: 3 marks.
- Accurate one-line description of each device's capability: 3 marks.
- Generation and defining technology named correctly for each computer device, justified by the technology rather than the date alone (Analyse): 3 marks.
- Display is clear and readable for a Class 6 audience: 1 mark.

Further reading

- Khan Academy. *Computers and the Internet* (free course developed with Code.org). <https://www.khanacademy.org/computing/computers-and-the->

internet

- Shelly, G. B., & Vermaat, M. E. *Discovering Computers*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 2. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>

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